MILITARY SPECIFICATION

ALTERNATOR, TURBINE (AIR DRIVEN)

This Specification is approved for use within the U.S. Army Armament Munitions and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements, quality assurance provisions, and packaging of the turbine alternator (T/A) for the M734 Multi-Option Mortar Fuze.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

MILITARY

MIL-P-116	- Preservation, Method of Quality Requirements and Inspection
MIL-A-48078	- Ammunition, Standard Quality Assurance Provisions, General Specification for

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, U.S. Army Armament, Munitions and Chemical Command, Attn. AMSMC-QA, Picatinny Arsenal, New Jersey 07806-5000 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-202	- Test Methods for Electronic and Electrical Component Parts
MIL-STD-331	- Fuze and Fuze Components, Environmental and Performance Tests
MIL-STD-454	- Standard General Requirement for Electric Equipment (Requirement 5, Except Paragraph 4.3)
MIL-STD-1168	- Ammunition Lot Numbering
MIL-STD-1169	- Packaging, Packing and Marking for Shipment of Inert Ammunition Components
MIL-STD-45662	- Calibration of Standards

2.1.2 Other Government documents, drawings and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

DRAWINGS

HARRY DIAMOND LABORATORIES

PRODUCT AND PACKAGING DRAWINGS

11722760	- Turbine alternator, air driven

INSPECTION EQUIPMENT DRAWINGS

11730644	- Test Vehicle
11730645	- Test Equipment

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

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2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Materials. Materials shall be in accordance with the applicable drawings and specifications.

3.2 First article inspection & comparison samples. This specification contains technical provisions for first article inspection. Requirements for the submission of first article samples by the contractor shall be specified in the contract.

3.2.1 Comparison samples. Fifty units or as otherwise specified by the Government, selected at random from each accepted lot, shall be shipped within two working days of lot acceptance to the Government technical agency designated by the procuring activity (6.6).

3.3 Turbine alternator assemblies. The turbine alternator assemblies and parts shall comply with all requirements specified on drawing (dwg.) 11722760 and with all requirements specified in applicable specifications and standards.

3.3.1 Operating requirements.

3.3.1.1 Low pressure. The turbine alternator when subjected to an input air pressure of .206-.006 psig shall operate at -50°F at a frequency of 1000 Hz minimum to 1350 Hz maximum and provide a minimum voltage of 19.4 volts rms across a 600 ohm load. At +145°F the electrical output shall be 1250 Hz minimum to 1500 Hz maximum and provide a minimum voltage of 20.8 volts rms across a 600 ohm load. At the temperature extremes the T/A shall operate within the specification for a minimum of 10 seconds.

3.3.1.1.1 Low pressure parameter adjustment. After final assembly of the turbine alternator, it shall be subjected to an input air pressure of .206 - .006 psig and ambient temperature. The turbine alternator shall be adjusted under above conditions until it operates at a frequency of 1250 Hz minimum to 1450 Hz maximum and provide a minimum of 20.8 volts rms across a 600 ohm load.

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3.3.1.2 High pressure. The turbine alternator when subjected to an input air pressure of 3.5 psig shall operate, over all temperature extremes, at a maximum frequency of 5900 Hz and provide a minimum of 21.0 volts rms (averaged over a minimum of one minute operation time) across a 600 ohm load. A gradual increase or decrease in frequency or voltage during the one minute operation is allowable but at no time shall the voltage drop below the minimum required in the specification. During the one minute operation, there shall be no sudden decrease or increase in operational characteristics.

3.3.1.3 Temperature requirements. The alternator shall meet the requirements of (3.3.1.1) and (3.3.1.2) when tested at temperatures of -50°F minimum to +145°F maximum.

3.3.1.4 Rotation. The turbine magnet/shaft assembly shall rotate within the housing and end plate with only magnet coupling resistance, free of physical rubbing contact friction (except for bearing component friction).

3.3.2 Coil Resistance. The direct current resistance of the coil winding shall be 57.5 ± 3.0 ohms.

3.3.3 Temperature and humidity. The alternator assembly (11722760), assembled in a fixture similar to an inert M734 fuze, shall meet performance requirements (3.3) after being subjected to temperature and humidity conditioning in accordance with MIL-STD-331A, test 105.1, procedure 2.

3.4 Soldering. Soldering shall be performed in accordance with MIL-STD-454, procedure 5, except for 4.3.

3.5 Lubrication. The bearing lubricant mixture shall not fall4below 7:1 due to xylene evaporation or other factor nor shall it exceed a ratio of 9.5:1.

3.6 Magnet to shaft pull test (Destructive test). The magnet shall not move on the shaft when an axial force of 120 pounds is applied for 10 seconds minimum.

3.6.1 Magnet crack test. The magnet (dwg. 11722761) shall not lose its physical integrity when being spun at a speed of 140,000 rpm minimum while being held in an approved device.

3.7 Data requirements. The contractor shall generate data in accordance with the requirements of the data item description cited in 6.5.

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3.8 Tolerances. The following table of Tolerances shall be applied where applicable to specified test conditions or measurements, not to the equipment used for these measurements unless otherwise defined in the operating requirements.

a.	Temperature	±2°F
b.	Force, Weight or Torque	±5%
c.	Resistance	±2%
d.	Frequency	+2%
e	Pressure, low	-3%
f.	Pressure, High	+1%
g	Voltage	±1%

3.9 Workmanship. Parts and assemblies shall be free from chips, dirt, grease and other foreign matter that may impair proper functioning. The cleaning method used shall not be injurious to any of the parts, nor shall the parts be contaminated.

3.9.1 Bearing radii surface. The bearing radii surface, which forms the bearing inner races, shall be free of any flats, pits, bumps, tool marks, or any other surface irregularities, in excess of the specified finish.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Unless otherwise specified herein or in the contract, the provisions of MIL-A-48078 shall apply and are hereby made a part of this detail specification.

4.1.1 Responsibility for compliance. All items must meet all requirements of section 3 and 5. The inspections set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirement in the specification shall not relieve the contractor of

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the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Classification of inspections. The following types of inspections shall be conducted on this item:

a. First article inspection

b. Quality conformance inspection

4.3 First article inspection.

4.3.1 Submission. The contractor shall submit a first article sample as designated by the Contracting Officer for evaluation in accordance with provisions of 4.3.2. The first article sample shall consist of the following items in sample quantities as indicated.

Part Description	Drawing Number	Quantity
Alternator Assembly	11722760	294
Coil Assembly	11722766	10
Bobbin and Contact Assembly	11722765	10
Contact	11722764	10
Bobbin	11722763	10
Magnet Shaft and Washer Assembly	11730640	10
Shaft	11730641	10
Magnet	11722761	10
Housing	11730642	10
End Plate	11730643	10
Housing & Bearing Assembly	11723110	10
End Plate & Bearing Assembly	11723111	10
Bearing	11730638	10
Turbine, 10 Blade	11722762	10

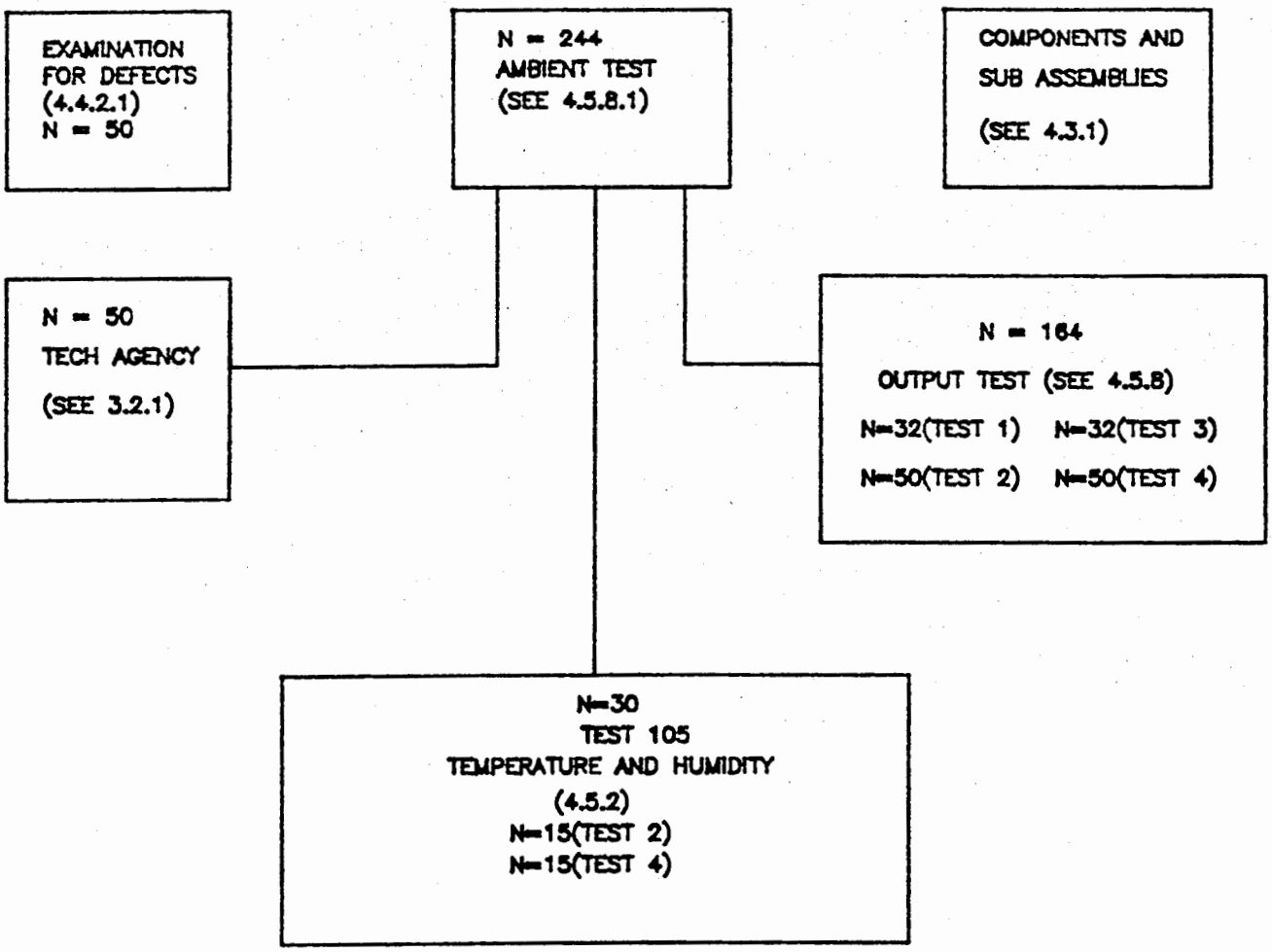
NOTE: If any part or component is fabricated from a multicavity mold process, 10 sets of parts are required from each cavity for examination under the appropriate paragraph.

4.3.2 Inspections to be performed. See MIL-A-48078, Figure 1 and Table I specfied herein.

4.3.3 Rejection. See MIL-A-48078.

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TURBINE ALTERNATOR (AIR DRIVEN)
FIRST ARTICLE SAMPLE FLOW DIAGRAM



OPERATIONAL TEST (OP) – SEE 4.5.8

1 – LOW PRESSURE
LOW TEMPERATURE
AC 1, RE 2

2 – HIGH PRESSURE
LOW TEMPERATURE
AC 1, RE 2

3 – LOW PRESSURE
HIGH TEMPERATURE
AC 1, RE 2

4 – HIGH PRESSURE
HIGH TEMPERATURE
AC 1, RE 2

FIGURE 1

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TABLE I - First article inspection

MIL-A-63991(AR)

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 3		DRAWING NUMBER
	Alternator, Turbine (Air Driven)				See Below
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Alternator Assembly (dwg. 11722760)				
	Examination for defects	50		3.3	4.4.2.1
	Alternator output test	244		3.3.1	4.5.8
	Turbine to shaft pull test	50	1/	3.3	4.5.7
	Temperature and humidity test	30		3.3.3	4.5.2
	Coil Assembly (dwg. 11722766)				
	Examination for defects	10		3.3	4.4.2.2
	Bobbin and Contact Assembly (dwg. 11722765)				
	Examination for defects	10		3.3	4.4.2.3
	Contact (dwg. 11722764)				
	Examination for defects	10		3.3	4.4.2.4
	Bobbin (dwg. 11722763)				
	Examination for defects	10		3.3	4.4.2.5

NOTES:
1/ The turbine to shaft pull test may be performed on the same units after they have been subjected to the examination for defects.

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TABLE I - First article inspection

CLASSIFICATION OF DEFECTS & TESTS MIL-A-63991(AR)

PARAGRAPH	TITLE		SHEET 2 OF 3		DRAWING NUMBER
	Alternator, Turbine (Air Driven)				See Below
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Magnet Shaft and Washer Assembly (dwg. 11730640)				
	Examination for defects	10		3.3	4.4.2.6
	Magnet to shaft pull test	10		3.6	4.5.3
	Shaft (dwg. 11730641)				
	Examination for defects	10		3.3	4.4.2.7
	Magnet (dwg. 11722761)				
	Examination for defects	10		3.3	4.4.2.8
	Magnet cracked test	10		3.1	4.5.4
	Housing (dwg. 11730642)				
	Examination for defects	10		3.3	4.4.2.9
	End Plate (dwg. 11730643)				
	Examination for defects	10		3.3	4.4.2.10
	Housing & Bearing Assembly (dwg. 11723110)				
	Examination for defects	10		3.3	4.4.2.11

NOTES:

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TABLE I - First article inspection

CLASSIFICATION OF DEFECTS & TESTS

MIL-A-63991(AR)

PARAGRAPH	TITLE		SHEET 3 OF 3		DRAWING NUMBER
	Alternator, Turbing (Air Driven)				See Below
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	End Plate & Bearing Assembly (dwg. 11723111) Examination for defects	10		3.3	4.4.2.12
	Bearing (dwg. 11730638) Examination for defects	10		3.3	4.4.2.14
	Turbine, 10 Blade (dwg. 11722762) Examination for defects	10		3.3	4.4.2.15

NOTES:

AMSMC Form 1570, 1 Feb 85 Replaces DRSMC-OA (D) Form 160, 1 Aug 83, which may not be used

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4.4 Quality conformance inspection.

4.4.1 Inspection lot formation. Inspection lots shall comply with the lot formation provisions of MIL-A-48078. In addition, inspection lots of turbine alternators shall meet the following criteria:

a. A lot shall consist of alternators produced at one manufacturing location within one working month.

b. Lot size shall not exceed 50,000 units.

c. Design or process changes are not allowed during the production of any praticular alternator lot.

4.4.2 Examination. See MIL-A-48078.

a. Sampling plans. Unless otherwise specified in the Classification of Defects and Test tables, sampling plans and procedures for major and minor defects shall be in accordance with MIL-STD-105, Inspection Level II.

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.1	Alternator, Turbine (Air Driven)				11722760
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Alternator output test		1.0% (Level S-4)	3.3.1	See Note
102	Improper shaft end play		0.40%	3.3	Gage
103	Improper assembly of turbine to shaft		0.40%	3.3	4.5.7
104	Distance from top of turbine to top of end plate surface		0.40%	3.3	Gage
105	End plate improperly secured to housing		0.40%	3.3	Visual/Gage
106	Free movement of turbine magnet/shaft assembly		0.40%	3.3.1.4	Tactile/Manual
107	.152 Dimension out-of-tolerance		0.40%	3.3	Gage
108	Weld above housing edge		0.40%	3.3	Gage
109	Parameter adjustment		100%	3.3.1.1.1	Test/4.5.8.1.
110	Maximum dimension from bottom of housing to top of end plate (.358)		0.40%	3.3	Gage
Minor					
201	Height of contact		0.65%	3.3	Gage
202	Position of contact		0.65%	3.3	Gage
203	Evidence of poor workmanship		0.65%	3.9	Visual

NOTES: Inspections level shall be applied to 4.5.8.2 and 4.5.8.5 individually.

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QUALITY CONFORMANCE INSPECTION

MIL-A-63991(AR)

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.2	Coil Assembly				11722766
					NEXT HIGHER ASSEMBLY 11722760
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Improper attachment of wire to contact (two places)		0.40%	3.3	Visual
102	Improper resistance between contacts		0.40%	3.3.2	4.5.9/Test
103	Improper insulation of contact legs		0.40%	3.3	Visual
Minor					
201	Evidence of poor workmanship		0.65%	3.9	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 11722765
4.4.2.3	Bobbin and Contact Assembly				NEXT HIGHER ASSEMBLY 11722766
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Minor					
201	Contacts improperly heat staked(4 places)		0.65%	3.3	Visual/Manual
202	Contact not in full face contact with bobbin		0.65%	3.3	Visual
203	Evidence of poor workmamship		0.65%	3.9	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 11722764
4.4.2.4	Contact				NEXT HIGHER ASSEMBLY 11722765
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Location of .063 dimension out of tolerance		0.40%	3.3	Gage
102	Outer interface contact height		0.40%	3.3	Gage
Minor					
201	Diameter of holes		0.65%	3.3	Gage
202	Location of holes		0.65%	3.3	Gage
203	Evidence of poor workmanship		0.65%	3.9	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-A-63991(AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 11722763
4.4.2.5	Bobbin				NEXT HIGHER ASSEMBLY 11722765
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Location of mounting posts incorrect	50		3.3	Gage
102	Location of contact studs incorrect	50		3.3	Gage
Minor					
201	Distance between outside faces	50		3.3	Gage
202	Boss diameter	50		3.3	Gage
203	Inside diameter	50		3.3	Gage
204	Evidence of poor workmanship			3.9	Visual

NOTES: 1/ The same 50 samples shall be used for all examinations.
2/ A new sample shall be drawn from each new lot, molding run or cavity.
3/ If two or more defects are found, the turbine shall be rejected or screened in its entirety for the defect(s) and be resubmitted for inspection.

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.6	Magnet Shaft & Washer Assembly	1 OF 1	11730640
			NEXT HIGHER ASSEMBLY 11722760

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Runout between magnet and shaft (2 axis)		0.40%	3.3	Gage
102	Runout of magnet faces to shaft		0.40%	3.3	Gage
103	Distance from aft shoulder of bearing radius to aft face of washer		0.40%	3.3	Gage
Minor					
201	Distance from aft shoulder of bearing radius to aft face of magnet (.073)		0.65%	3.3	Gage
202	Evidence of surface defects on raceways and datum A surfaces		0.65%	3.3	Visual
203	Magnet to shaft pull test		0.65%	3.6	4.5.3
204	Evidence of poor workmanship		0.65%	3.9	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 11730641
4.4.2.7	Shaft				NEXT HIGHER ASSEMBLY 11730640
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Surface of bearing radii finish		0.40%	3.8.1	Gage
102	Length of aft shaft from end to first bearing ridge		0.40%	3.3	Gage
103	Length from bearing ridge to bearing ridge		0.40%	3.3	Gage
104	Bearing surface diameter		0.40%	3.3	Gage
105	Largest diameter of bearing race (.1105 dimension)		0.40%	3.3	Gage
106	Improper angle cut of aft "V" slot		0.40%	3.3	Gage
Minor					
201	Improper width of aft "V" slot		0.65%	3.3	Gage
202	Evidence of poor workmanship		0.65%	3.9	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET	OF	DRAWING NUMBER
4.4.2.8	Magnet	1	1	11722761

NEXT HIGHER ASSEMBLY: 11730640

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Magnet cracked		0.40%	3.3	4.5.4/Test
102	Outside diameter		0.40%	3.3	Gage
103	Perpendicularity of outside diameter magnet face		0.40%	3.3	Gage
104	Parallelism of magnet faces		0.40%	3.3	Gage
Minor					
201	True position of fill cavities to outside diameter (.293)		0.65%	3.3	Gage
202	True position of fill cavity relief to outside diameter (.237)		0.65%	3.3	Gage
203	True position of inside diameter to outside diameter (.174)		0.65%	3.3	Gage
204	Evidence of poor workmanship		0.65%	3.9	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 11730642
4.4.2.9	Housing				NEXT HIGHER ASSEMBLY 11723110
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Inside diameter of bearing cavity(.1743)	50		3.3	Gage
102	Outside diameter of bearing cavity(.2055)	50		3.3	Gage
103	Height of bearing cup face from housing face (.355)	50		3.3	Gage
104	Height of bearing cup face from end plate housing shoulder (.0685)			3.3	Gage
105	Inner housing shelf diameter (1.128)				
Minor					
201	Inside flange leg diameter	50		3.3	Gage
202	Inside diameter of bearing shaft cavity	50		3.3	Gage
203	MMC of cutouts incorrect	50		3.3	Gage
204	Height bearing cavity boss to cup face	50		3.3	Gage
205	Evidence of poor workmanship		0.65%	3.9	Visual

NOTES: 1/ The same 50 samples shall be used for all examinations. 2/ A new sample shall be drawn from each new lot, stamping run or fixture. 3/ If two or more defects are found, the housing lot shall be rejected or screened in its entirety for the defect(s) and be resubmitted for inspection.

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.10	End Plate		11730643
			NEXT HIGHER ASSEMBLY 11723110

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Inside diameter of bearing cavity (.1743)	50		3.3	Gage
102	Outside diameter of end plate (1.130)	50		3.3	Gage
103	Depth of bearing cavity (.047)	50		3.3	Gage
Minor					
201	Diameter of shaft hole	50		3.3	Gage
202	Inside flange leg diameter	50		3.3	Gage
203	MMC of slots incorrect	50		3.3	Gage
204	Evidence of poor workmanship		0.65%	3.9	Visual

NOTES: 1/ The same 50 samples shall be used for all examinations. 2/ A new sample shall be drawn from each new lot, stamping run or fixture. 3/ If two or more defects are found, the end plate lot shall be rejected or screened in its entirety for the defect(s) and be resubmitted for inspection.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-A-63991(AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 11723110
4.4.2.11	Housing & Bearing Assembly				NEXT HIGHER ASSEMBLY 11722760
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Bearing balls not free to rotate		100%	3.3	Tactile/Manual/4.5.10
102	Inside diameter of bearing		0.40%	3.3	Gage
103	Outside diameter of bearing boss		0.40%	3.3	Gage
Minor					
201	Evidence of poor workmanship		0.65%	3.9	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.12	End Plate & Bearing Assembly				11723111
					NEXT HIGHER ASSEMBLY 11722760
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Bearing balls not free to rotate		100%	3.3	Manual/4.5.10
102	Exceeds inner ball circle diameter		0.40%	3.3	Gage
Minor					
201	Evidence of poor workmanship		0.65%	3.9	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET	OF	DRAWING NUMBER
4.4.2.13	Lubricant Mixture, Bearing	1	1	11723115
				NEXT HIGHER ASSEMBLY 11730638

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	As-mixed ratio out-of-tolerance	each batch		3.3	4.5.5/Test
102	Ratio in useage below minimum ratio	each lubricated bearing lot		3.5	4.5.6/Test
103	Ratio in useage above maximum ratio	each lubricated bearing lot		3.5	4.5.6/Test
Minor	None defined				

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.14	Bearing				11730638
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Inside diameter of ball circle		0.40%	3.3	Gage
102	Number of balls incorrect		0.40%	3.3	Visual
103	Outside diameter of race		0.40%	3.3	Gage
Minor					
201	Inside diamter of race (forward .115)		0.65%	3.3	Gage
202	Inside diameter of race (aft .1065)		0.65%	3.3	Gage
203	Evidence of poor workmanship		0.65%	3.9.1	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

MIL-A-63991(AR)

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.15	Turbine, 10 Blade		11722762
			NEXT HIGHER ASSEMBLY 11722760

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Diameter of shaft hole	50		3.3	Gage
Minor					
201	Diameter of turbine base	50		3.3	Gage
202	Height of turbine fins	50		3.3	Gage
203	Evidence of poor workmanship		0.65%	3.9	Visual

NOTES: 1/ The same 50 samples shall be used for all examinations. 2/ A new sample shall be drawn from each new lot, molding run or mold cavity. 3/ If two or more defects are found, the turbine shall be rejected or screened in its entirety for the defect(s) and be resubmitted for inspection.

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4.4.3 Testing. See 4.5

4.4.4 Inspection equipment. The inspection equipment required to perform the examinations and test prescribed herein is described in the "Paragraph Reference/Inspection Method" column in the tables starting with paragraph 4.4.2.1. Calibration procedures shall conform to MIL-STD-45662. The contractor shall submit for approval inspection equipment designs in accordance with the terms of the contract. See Section 6 of MIL-A-48078 and 6.3 herein.

4.5 Methods of inspection.

4.5.1 Materials, components and processes. Compliance with all requirements of Section 3 of this specification shall be ascertained by current and continuing examination of inspection and test data to determine that all components (parts, subassemblies, and materials) have been inspected and tested and found to comply with their respective drawing and specification requirements, and that all specified manufacturing processes have been followed.

4.5.1.1 Visual and mechanical inspection. The visual, manual, tactile and mechanical inspections and tests shall verify compliance with the requirements of Section 3 and 5 of this specification in accordance with 4.4.2 herein, Figure 1 and Table I.

4.5.2 Temperature humidity test. A sample of thirty alternators shall be selected, inserted into dummy M734 fuze bodies or other Government approved test vehicles and subjected to the test as specified in MIL-STD-331, procedure 105. The test unit shall then be disassembled and the alternator be tested in accordance with 4.5.8.4 and 4.5.8.5. Any alternator that does not meet the performance requirements of 3.3.1.2 shall be classified defective. See Figure 1.

4.5.3 Magnet to shaft pull test. The magnet and shaft assembly (dwg. 11730640) shall be held stationary and an axial force of 120 pounds shall be applied to the magnet for 10 seconds. Any magnet that separates or moves from the shaft shall be classified defective.

4.5.4 Magnet crack test. The magnet (dwg. 11722761) shall be spun at a speed of 140,000 rpm for 15 seconds while being held in an approved shaft-like fixture. Any magnet that loses its physical integrity in any way shall be classified defective.

4.5.5 Bearing lubricant mixture test. Weigh the evaporation dish to be used. Weigh a sample of mixture into the evaporation dish. The depth of lubricant shall be no greater than .010 inch. Place in an oven set at 180°F ± 10°F for two hours. Weigh the evaporating dish containing the residue and subtract the weight of

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the evaporation dish. The weight of the residue shall be between 9.52% and 10.53% of the original lubricant weight. If the above criteria are not met, the lubricant mixture has failed this test.

4.5.6 Bearing lubricant useage ratio test. Draw a sample of the mixture after it is used for bearing lubrication. Weigh the evaporation dish to be used. Weigh a sample of mixture in the evaporation dish. The depth of the lubricant shall be no greater than .010 inch. Place the sample in an oven set at 180°F ± 10°F for two hours. Weigh the evaporation dish containing the residue and substract the weight of the evaporation dish. The weight of the residue shall be between 9.52% and 12.5% of the original lubricant weight. If the above criteria are not met, the bearing lot lubricated with this mixture has failed this test.

4.5.7 Turbine to shaft pull test. The turbine alternator or shaft shall be held stationary and an axial pull force of 10 pounds shall be applied to the turbine. Any turbine that separates from the shaft shall be classified defective. This test is considered to be a destructive test. Note: even though this is a destructive test, turbines may be removed and replaced and units can be returned to the lot after removing and replacing the 10 blade turbine, 11722762, and retesting at ambient temperature in accordance with 4.5.8.1.

4.5.8 Alternator output test. The alternator shall be mounted in a government approved test set-up (see dwgs. 11730644 and 11730645) and subjected to the pressure and temperature conditions specified below. The alternator shall be conditioned at the required temperature for at least two hours. The test shall be initiated within 15 seconds after the alternator has been removed from the conditioning chamber.

4.5.8.1 Low pressure, ambient temperature test (parameter adjustment). The alternator shall be tested in accordance with 4.5.8 and shall be adjusted and shall meet the requirements of 3.3.1.1.1. Any alternator that does not meet the requirement shall be classed defective and shall not be returned to the lot.

4.5.8.2 Low-pressure, low temperature test. The alternator shall be tested in accordance with 4.5.8 and at -50°F shall meet the requirement of 3.3.1.1. Any alternator that does not meet the requirement shall be classified defective.

4.5.8.3 Low pressure, high temperature test. The alternator shall be tested in accordance with 4.5.8 and at +145°F shall meet the requirement of 3.3.1.1. Any alternator that does not meet the requirement shall be classified defective.

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4.5.8.4 High pressure, low temperature test. The alternator shall be tested in accordance with 4.5.8 and of -50°F shall meet the requirement of 3.3.1.2. Any alternator thatdoes not meet the requirement shall be classified defective.

4.5.8.5 High pressure high temperature test. The alternator shall be tested in accordance with 4.5.8 and at +145°F shall meet the requirement of 3.3.1.2. Any alternator that does not meet the requirement shall be classified defective.

4.5.9 Coil resistance. The direct current resistance of the coil winding shall be measured using method 303 of MIL-STD-202 and shall meet the requirements of 3.3.2.

4.5.10 Bearing Freeness. Carefully insert a needle gage between two balls of the bearing. Gently drive the ball complement for at least 90° of rotation, all 13 balls shall rotate in the raceway, any bearing that exhibits restraint of ball movement within the 90° shall be classed defective. This test shall be performed after the bearing ball complement has been size to .1022 as per drawing (11723110 and 11723111).

5. PACKAGING

5.1 Packing. Alternators shall be packaged in a clean economical container. The container shall limit the motion of the device within the container to 0.020 inch maximum in any direction, in accordance with MIL-P-116.

5.2 Marking. In addition to any special marking required by the contract, unit packages, intermediate packages, and shipping containers shall be marked in accordance with the requirements of MIL-STD-1169.

5.3 Records

5.3.1 Lot numbering. Lot numbering shall be as specified in MIL-STD-1168.

6. NOTES

6.1 Intended use. The turbine alternator, air driven, is designed to provide a high reliability source of electrical power for electronics and mechanical power for arming in fuze, mortar: multi-option, M734.

6.2 Ordering data. See MIL-A-48078.

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6.3 Submission of designs for approval. See 6.2.3 of MIL-A-48078. Submit equipment designs, as required to: Commander, U.S. Army Armament Research, Development and Engineering Center, ATTN: AMSMC-QAT-I(D), Picatinny Arsenal, NJ 07806-5000. This address will be specified on the Contract Data Requirements List, DD Form 1423 in the contract. Unless otherwise specified, data item DI-R-1714 will apply.

6.4 M734 dummy fuze. The prime contract or procuring agency will provide M734 dummy fuzes for the environmental test.

6.5 Data requirements. When this specification is used in an acquisition and data are required to be delivered, the data requirements identified below shall be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved Contract Data Requirements List (CDRL), incorporated into the contract. When the provisions of DOD FAR Supplement, Part 27, Sub-Part 27.410-6 (DD Form 1423) are invoked and the DD Form 1423 is not used, the data specified below shall be delivered by the contractor in accordance with the contract or purchase order requirements. Deliverable data required by this specification are cited in the following paragraphs.

Paragraph No.	Data Requirment	Applicable DID No.
4.3.2	First Article Inspection Report	DI-T-4902/M
4.4.2	Lot Summary Inspection Report	DI-R-1757
4.1	Quality Engineering Acceptance Inspection Equipment Descriptive Documentation	DI-R-1714

(Copies of data item descriptions required by the contractor in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

6.6 Government testing of comparison samples. These samples are subject to periodic retesting by the Government to insure that the first article approval requirements and test equipment certification continue to be met. Failure of these samples will not affect the acceptance of the lot but may indicate future mandatory action.

6.7 Alternate design. The only Government approved source for the turbine alternator as specified by MIL-A-63991 is:

Alinabal, Inc.
28 Woodmont Rd.
Milford, CT 06460

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The contractor may submit for Government approval an alternate design to the bearing used in the turbine alternator. The bearing must be of a type that is domestically produced or commercially available precision bearing in accordance with FAR 52.208-7000.

6.7.1 Performance requirements. The contractor must demonstrate to the approving agency that the design meets the input, output and interface requirements of 3.3.1 and 11272760.

6.7.2 Interface. The contractor must demonstrate to the approving agency that the design will not affect the interface requirements in its intended application of the turbine alternator (11722760).

6.7.3 Approval of alternate design. The Government technical agency is responsible for the approval of any alternate design. The contractor must receive a written approval from the Government before the design is incorporated in the M734 fuze assembly. The offeror shall submit as a minimum, a level III drawing package, sufficient sample data as required by first article samples, and performance requirements of section 3 and 4 of this specification.

The Government technical agency reserves the rights to subject any alternate design to any or all of the required physical or performance tests as specified in this specification. The technical agency is required to provide acceptance or rejection of the alternate design within 60 days after submission.

6.8 Subject term (key word) listing.

Alternator, turbine (air driven)
Fuze, multi-option: M734

Custodian:
Army-AR

Preparing activity:
Army-AR

(Project 1390-A574)

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